# PREDICTIVE ANALYSIS

# ACADEMIC YEAR 2025-26

# PROJECT – 1

# WATER QUALITY CLASSIFICATION FOR SAFE DRINKING PREDICTION

**GitHub:** https://github.com/BaintyKaur/Water-Quality-Classification-using-ML

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# PROBLEM DEFINITION

**The Challenge**

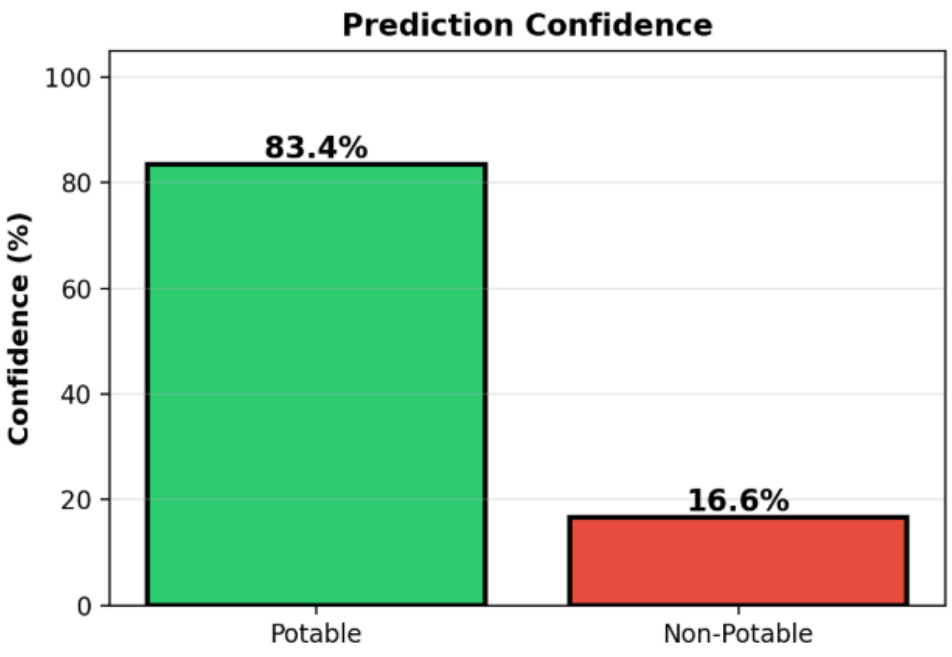
~ **2 billion** people consume contaminated water worldwide (WHO), leading to cholera, typhoid, dysentery, and more.

**Current Gaps**

- **Expensive** — lab equipment & trained staff
- **Slow** — results take days
- **Inaccessible** — rural & developing regions

**Our Solution**

- **AI-powered** binary classification (Safe / Not Safe)
- SMOTE + Hyperparameter Tuning + SHAP Explainability
- **Streamlit** deployment for real-time prediction



# LITERATURE REVIEW

| Feature | Description | Safe Limit |
|---|---|---|
| pH | Acidity/Basicity | 6.5 – 8.5 |
| Hardness | Ca/Mg Content | < 300 mg/L |
| Solids (TDS) | Dissolved Solids | < 500 ppm |
| Chloramines | Disinfectant Level | < 4 ppm |
| Sulfate | Sulfate Concentration | < 250 mg/L |
| Conductivity | Electrical Conductivity | < 400 µS/cm |
| Organic Carbon | Organic Matter | < 2 ppm |
| Trihalomethanes | THM Level | < 80 µg/L |
| Turbidity | Water Clarity | < 5 NTU |

**Study 1:**

Bhatt & Bhatt (2020)

Random Forest achieved **87% accuracy** on groundwater data

**Study2:**

Hameed et al. (2022)

XGBoost outperformed with **AUC = 0.91** & used SMOTE

**Study 3:**

Dawood et al. (2021)

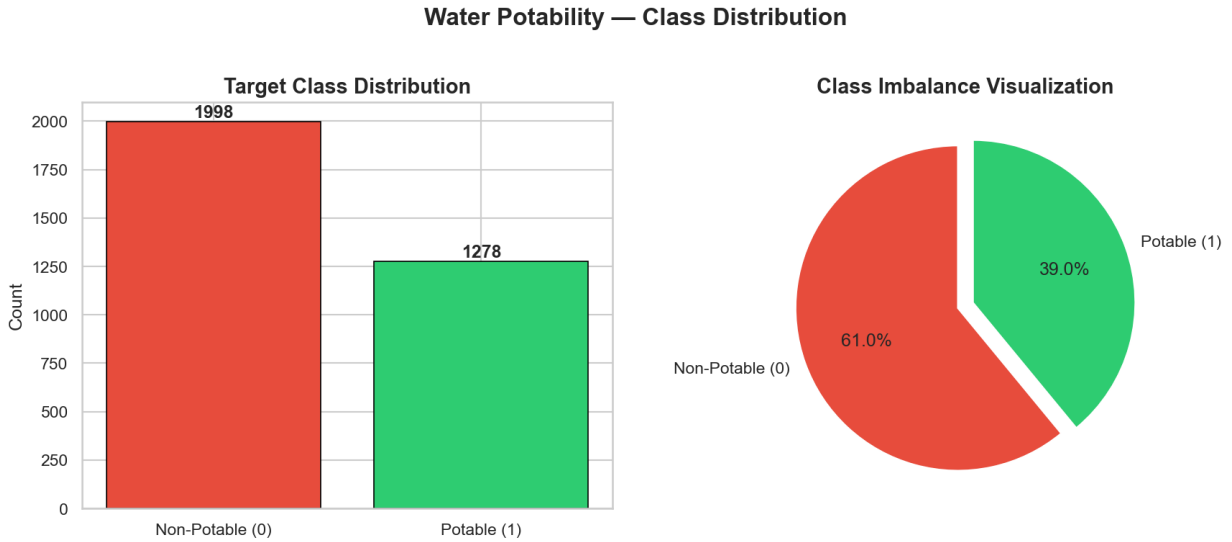
Classical ML (RF, XGB) match Deep Learning performance

# DATA COLLECTION & DATA UNDERSTANDING

| | |
|---|---|
| **NAME** | **Water Potability Dataset** |
| **SOURCE** | **Kaggle -** https://www.kaggle.com/datasets/adityakadiwal/water-potability - |
| **SIZE** | **3,276 rows × 10 columns** |
| **LICENSE** | **Open Database License (ODbL)** |

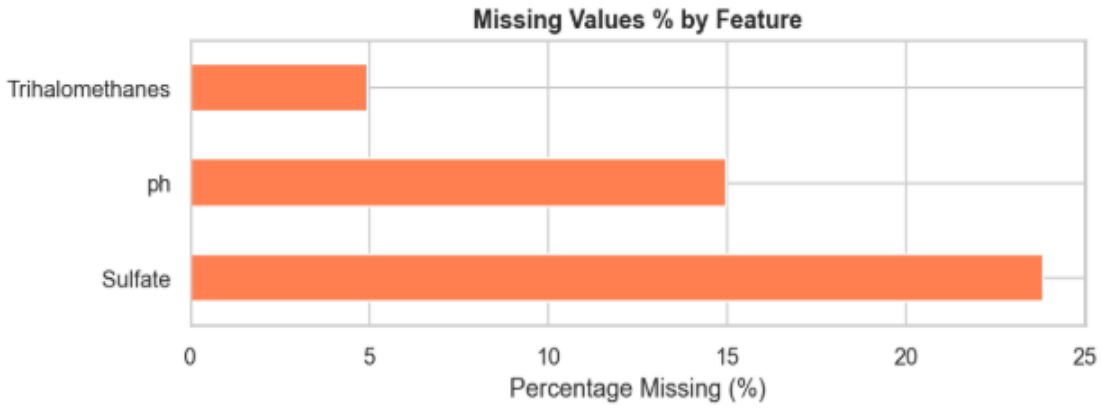
**Handling Strategy:**

- **Method:** Median imputation by class
- **Rationale:** Preserves class-specific distributions
- **Impact:** < 5% data loss after handling



# DATA PREPROCESSING & CLEANING

## I. Fills missing values using the median of each class separately (Non-Potable vs Potable)

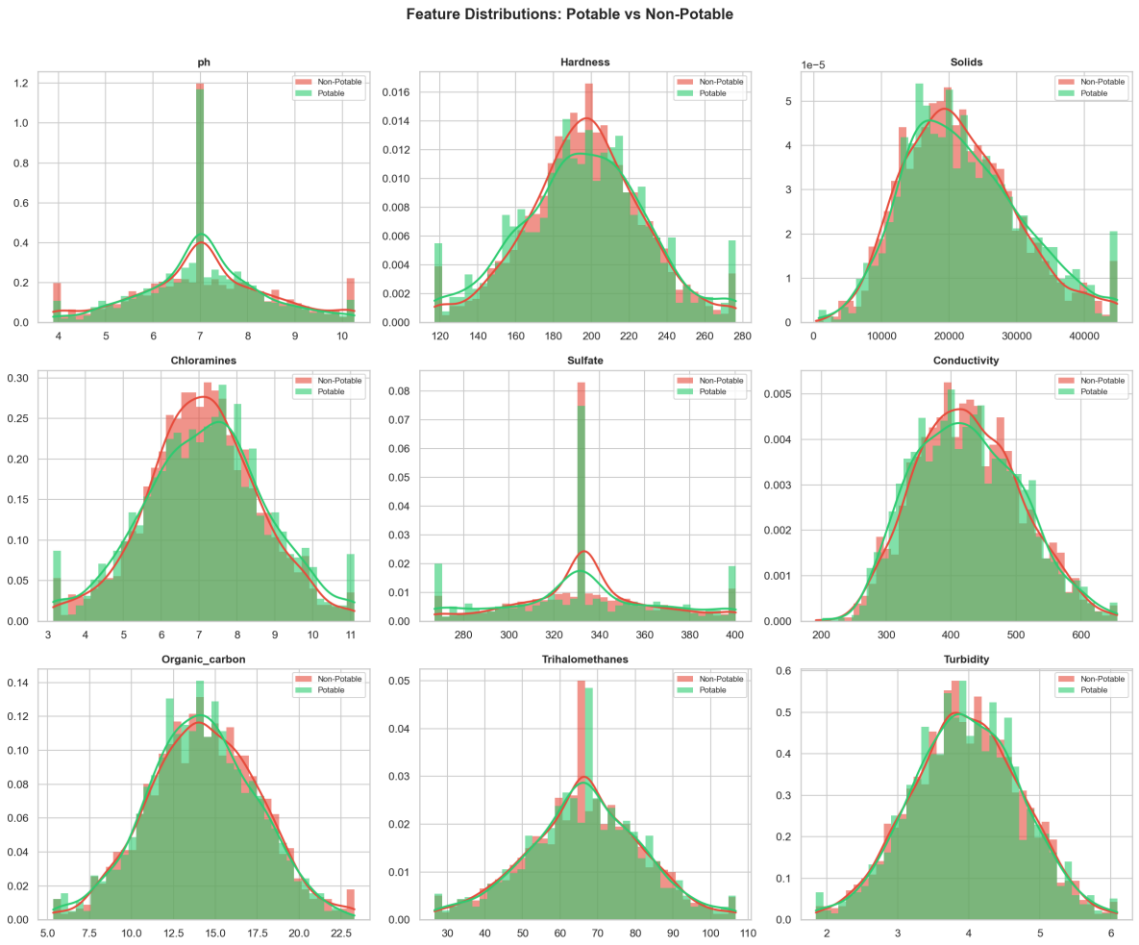



## II. Outlier Detection & Treatment

OUTLIER TREATMENT REPORT

| Feature | Lower Bound | Upper Bound | Outliers Found |
|---|---|---|---|
| ph | 3.889 | 10.259 | 142 |
| Hardness | 117.125 | 276.393 | 83 |
| Solids | -1832.417 | 44831.870 | 47 |
| Chloramines | 3.146 | 11.096 | 61 |
| Sulfate | 267.158 | 400.322 | 264 |
| Conductivity | 191.648 | 655.879 | 11 |
| Organic_carbon | 5.328 | 23.295 | 25 |
| Trihalomethanes | 26.619 | 106.695 | 54 |
| Turbidity | 1.849 | 6.091 | 19 |

# EXPLORATORY DATA ANALYSIS (EDA)




## Feature Distribution (Univariate Analysis)

- Most features are right-skewed (non-normal)
- High variability in Solids and Conductivity
- pH centred around neutral (~7)
- Turbidity shows low variation
- Outliers present in multiple features
- Outliers handled using IQR capping
- No single feature separates potable vs non-potable
- Data reflects real-world variability
- Requires multi-feature analysis
- Supports non-linear ML models

# FEATURE ENGINEERING & SELECTION

**Methods Used:**

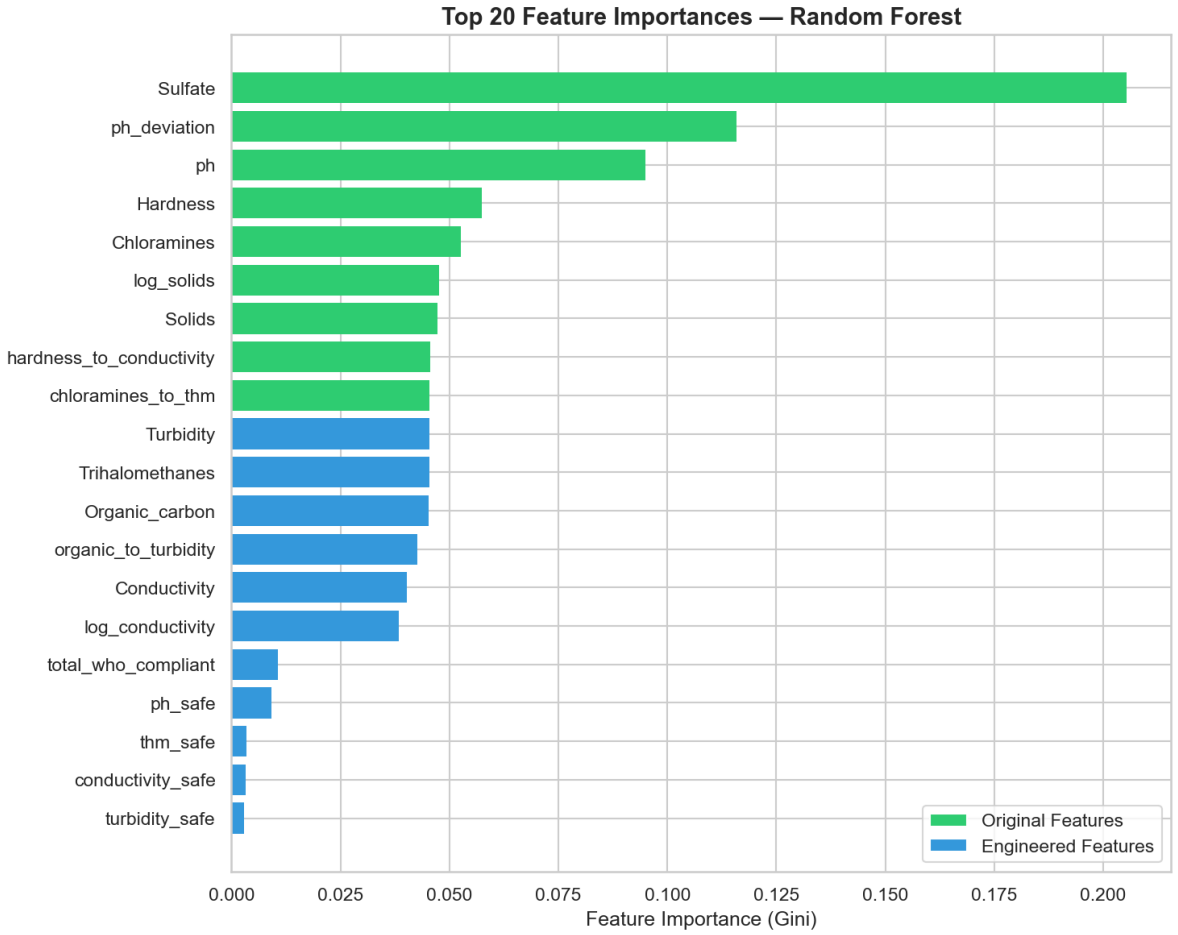
- Random Forest importance (Gini)
- Mutual Information scores
- Selection criterion: Importance > 0.01
- Result - 17 features selected from 28 total

**Feature Scaling:**

Method: StandardScaler (Zero-mean, unit variance)
Why: Required for algorithms like SVM and neural networks

**Train-Test Split**

- Split: 80-20 stratified
- Stratification: Preserves class distribution
- Result: Balanced train and test sets



# MODEL BUILDING & TRAINING

**Problem**: Dataset is imbalanced (more Non-Potable than Potable)

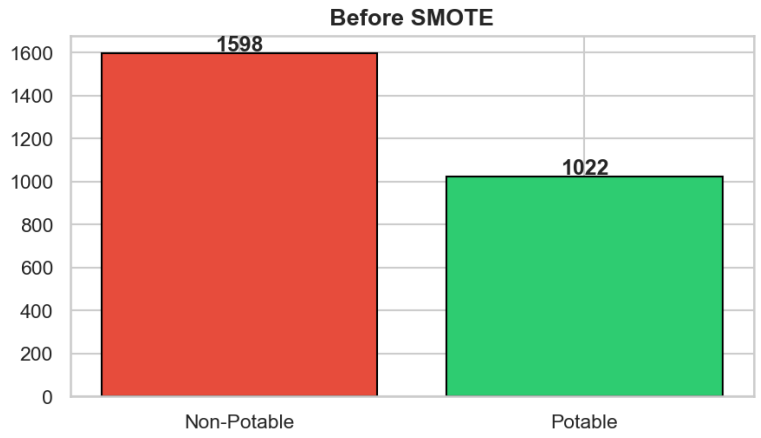
**Solution**: SMOTE (Synthetic Minority Over-sampling Technique)

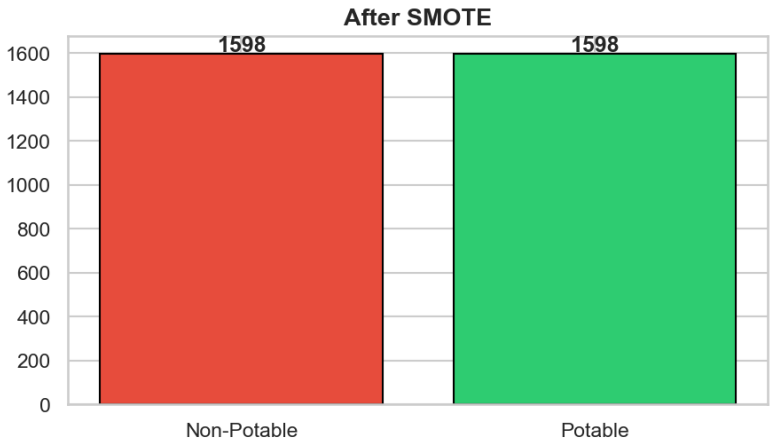
- Generates synthetic samples for the minority class (Potable)
- Applied **only on training data** (prevents data leakage)

**Result**:

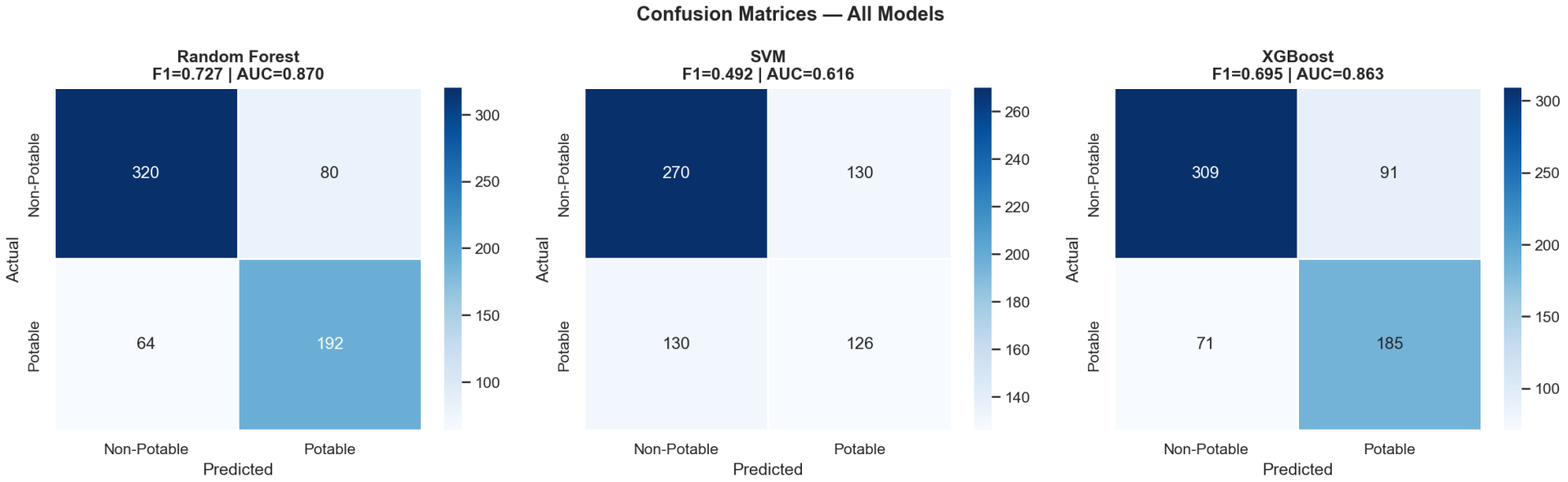
•Before SMOTE: ~1,800 Non-Potable, ~800 Potable (ratio: 2.3:1)

•After SMOTE: Perfect 1:1 balance (~1,600 each class)



# MODELS EVALUATION & COMPARISON




**Random Forest** – Best model
Correctly identified 320/400 unsafe + 192/256 safe samples
Only **64 false negatives** (unsafe called safe)
F1 = 0.727 | AUC = 0.870

**SVM** – Failed
**130 FN**— double RF's errors
Nearly coin-flip performance
F1 = 0.492 | AUC = 0.616

**XGBoost** — Close second
Similar performance but **71 false negatives** vs RF's 64
F1 = 0.695 | AUC = 0.863

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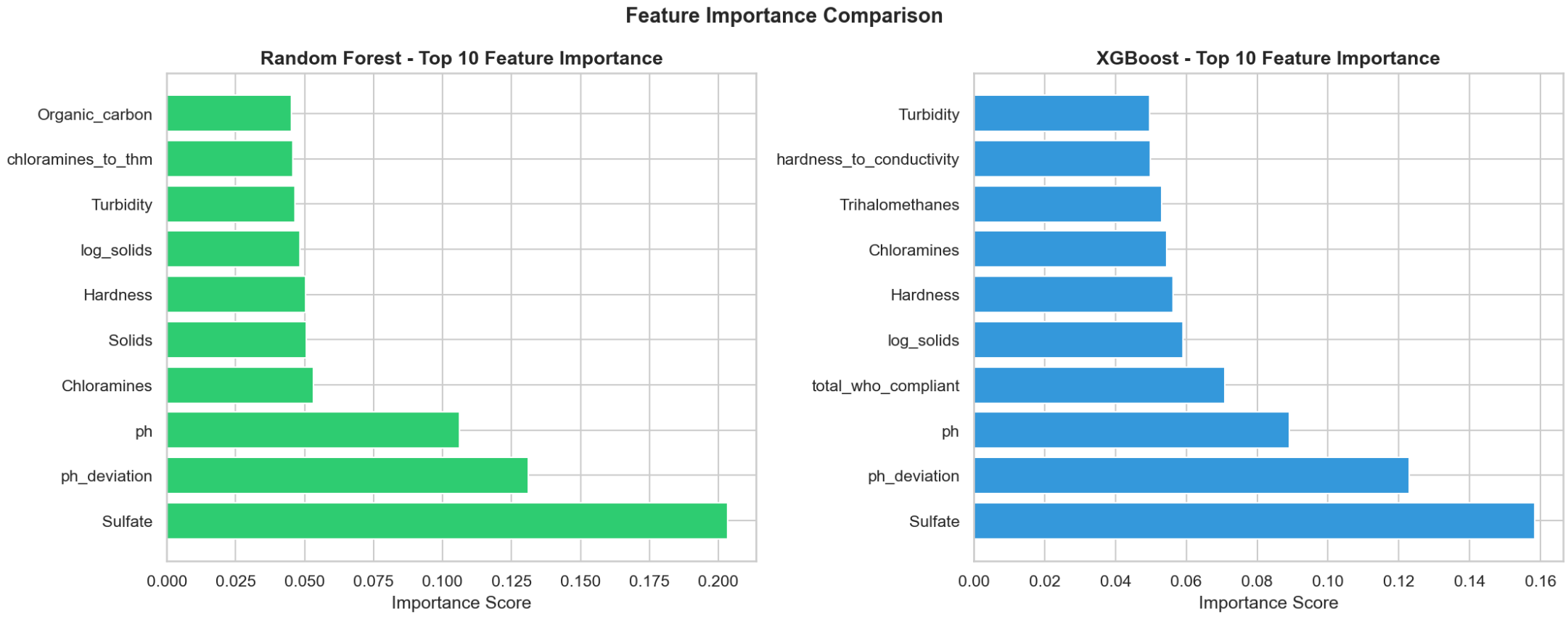

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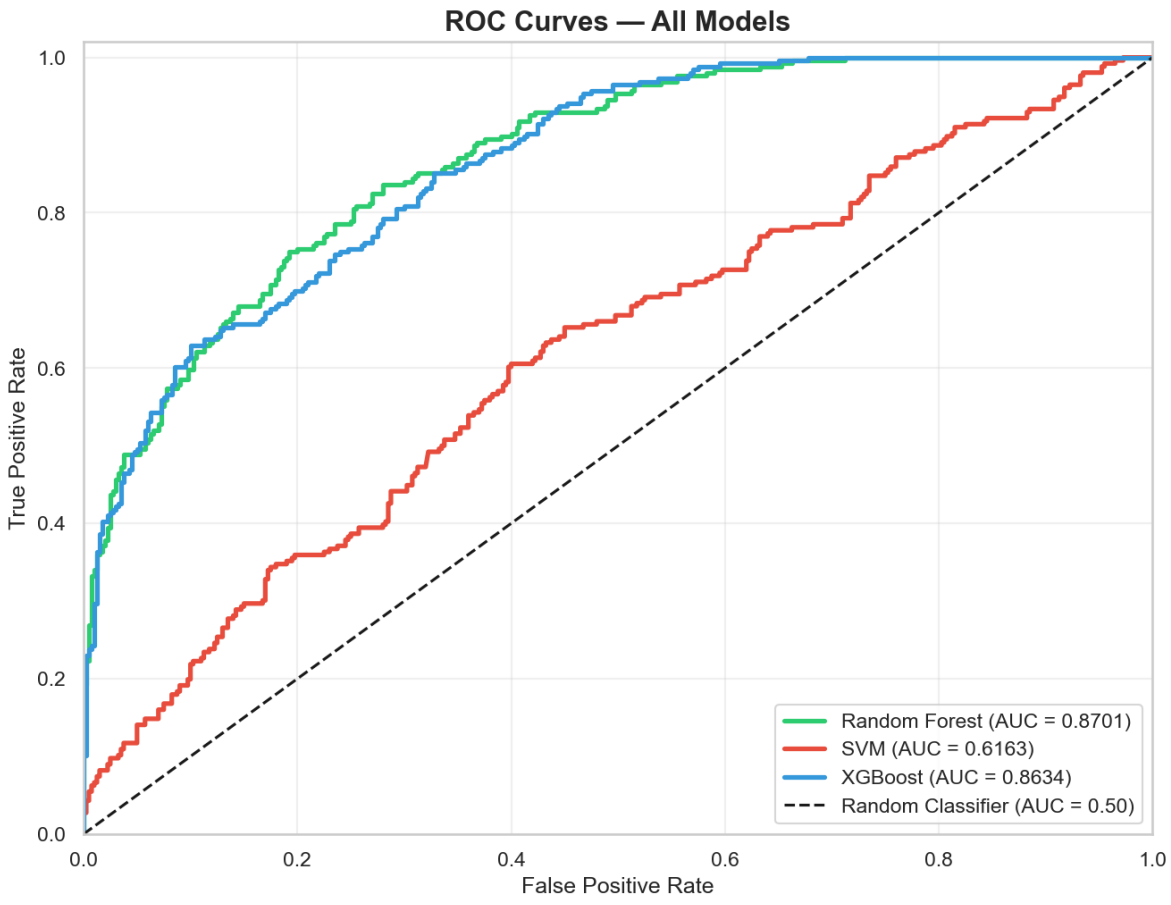
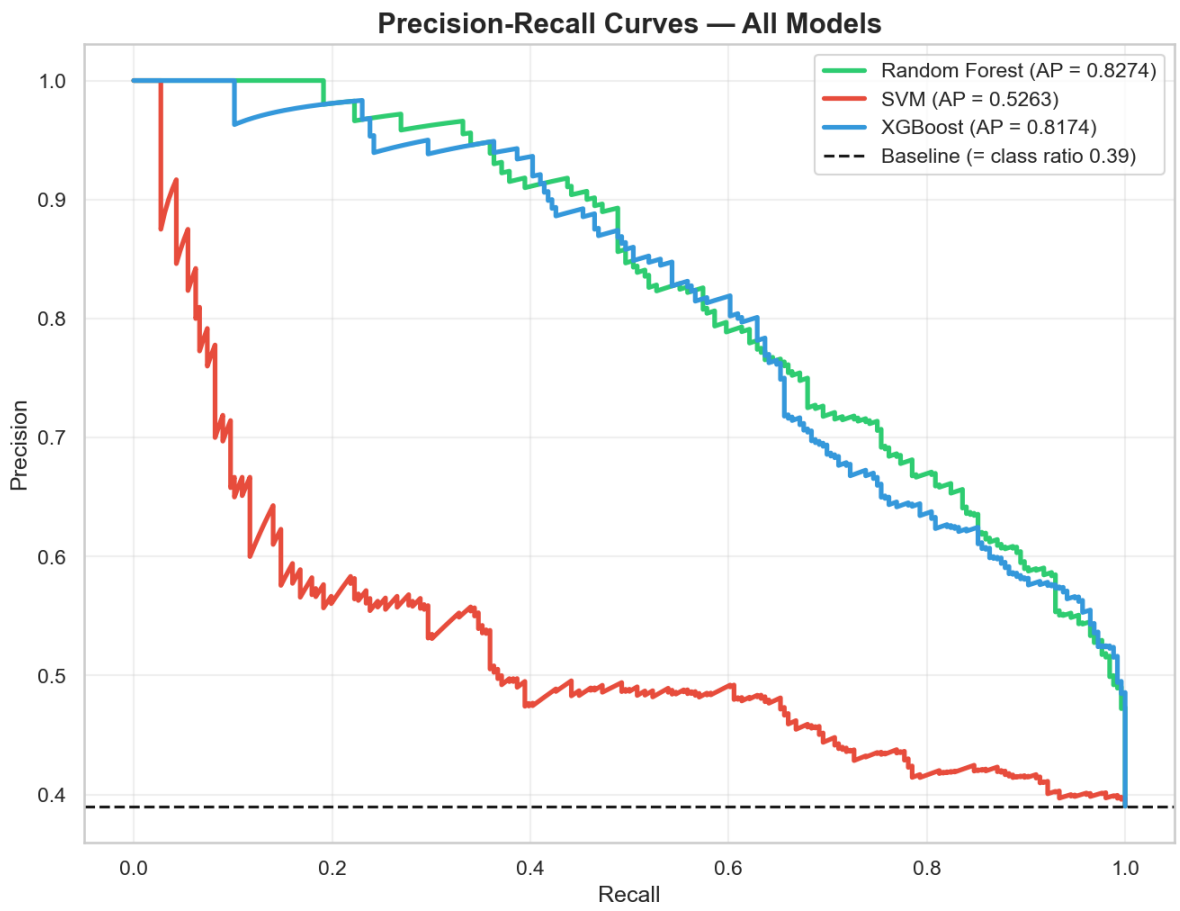
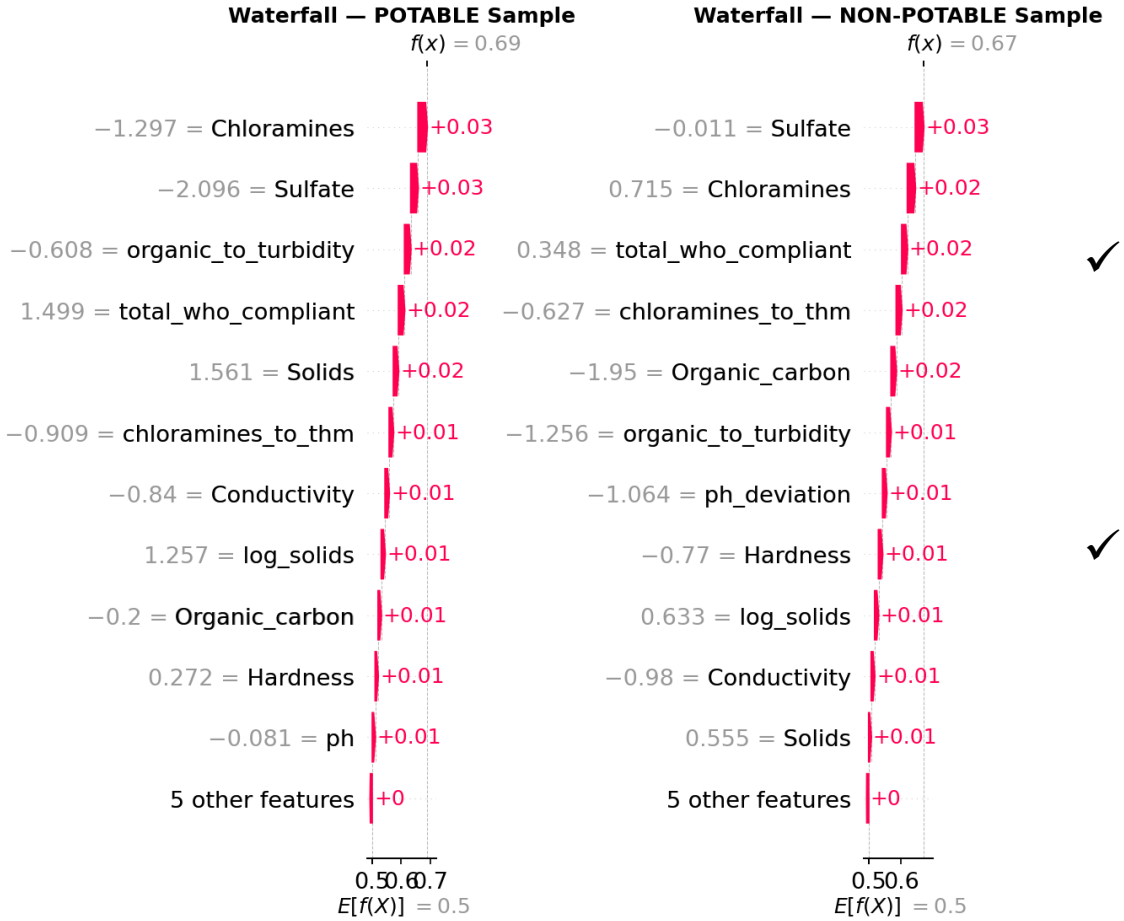


# MODEL INTERPRETATION & EXPLAINABILITY




- ✓ **Potable sample, f(x)=0.69:** Chloramines and Sulfate pushed the prediction above 0.5 toward potable. Multiple small positive contributions added up to 69% potable confidence
- ✓ **Non-potable sample, f(x)=0.67:** Sulfate again pushed toward potable — but notice the model still predicted 67% potable, which is a borderline case

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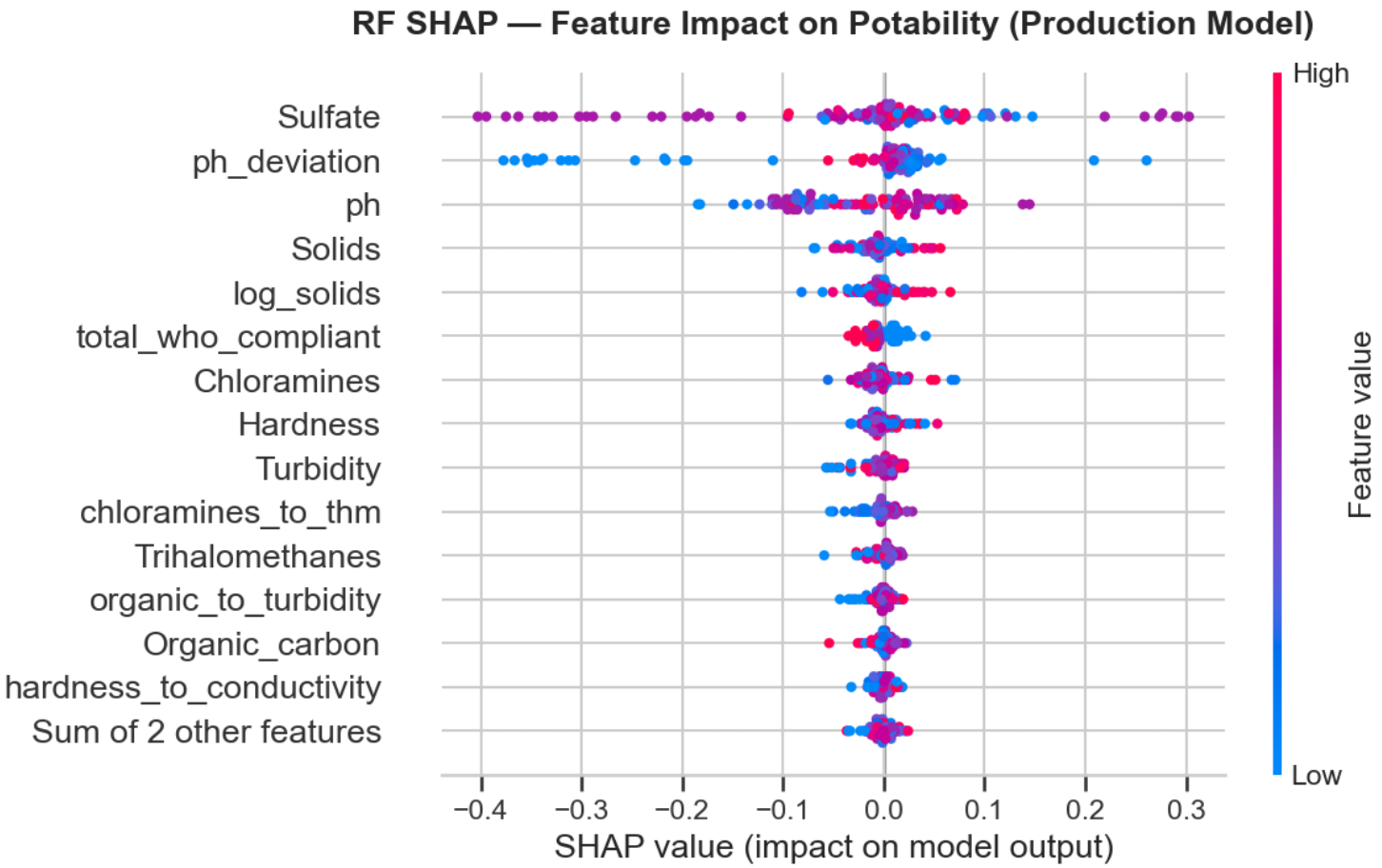

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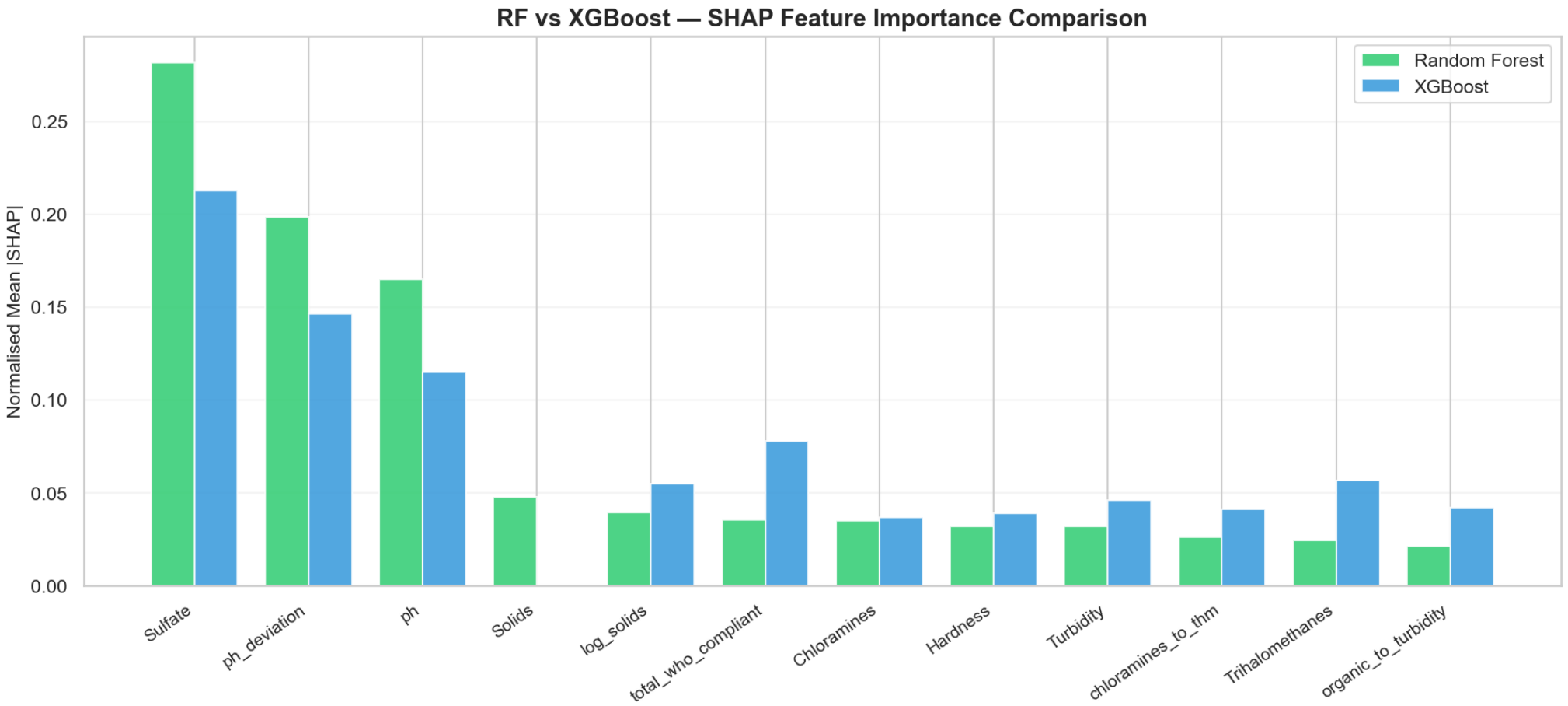

# THANK YOU!